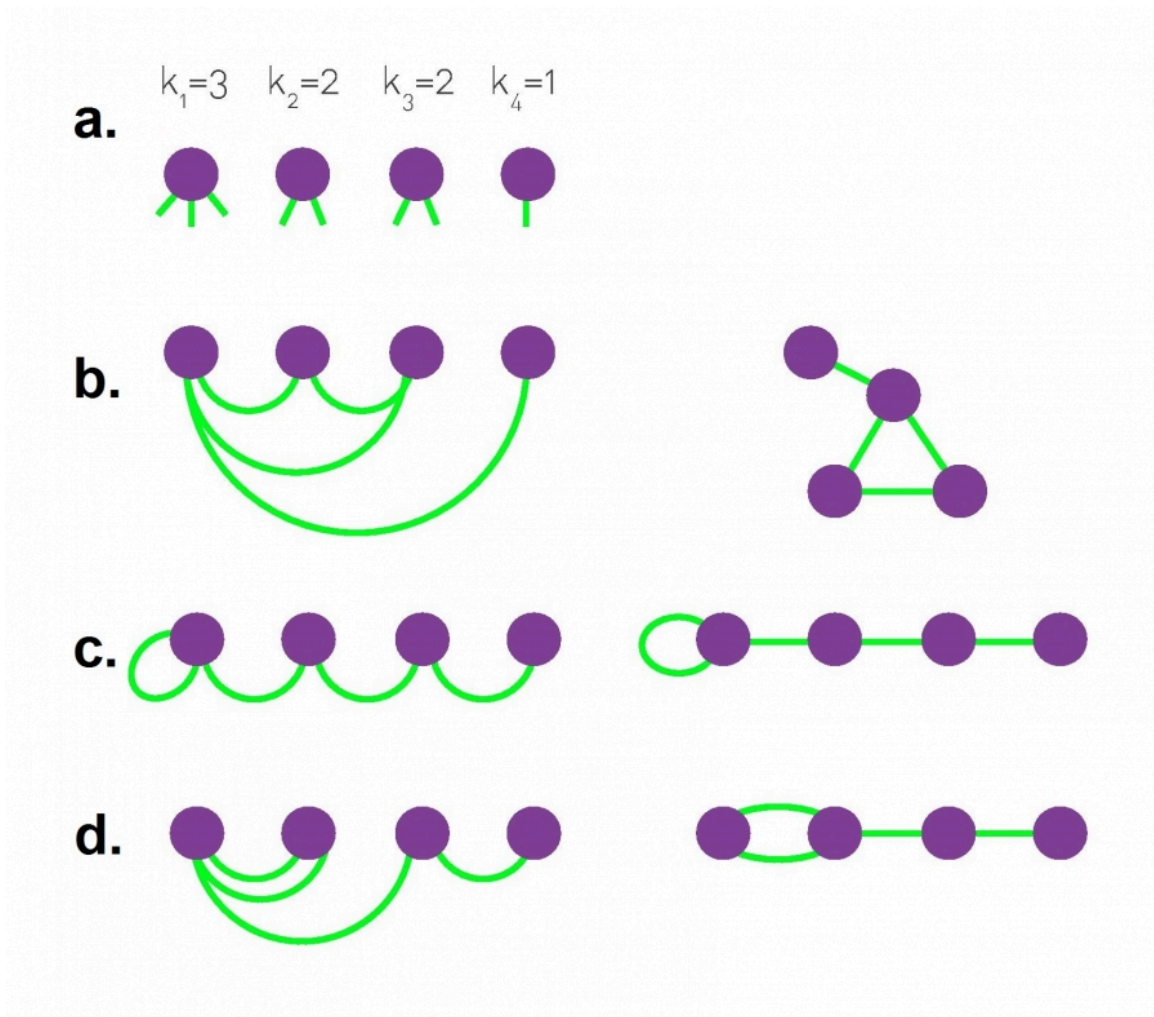


L4: Configuration Model

Suppose that a real-world communication network with one thousand nodes has a power-law degree distribution with exponent $\alpha = 2.5$ –you may want to investigate how this network will perform when it grows to ten thousand nodes, assuming that its degree distribution exponent remains the same.

How can we create a synthetic network that has a given degree distribution? This is a central question in network modeling.

A general way to create synthetic networks with a specified degree distribution p_k is the “configuration model”. The inputs to this model is **a)** the desired number of nodes n , and **b)** the degree k_i of each node i . The collection of all degrees specifies the degree distribution of the synthetic network.



The Configuration Model, Figure 4.15 from networksciencebook.com by Albert-László Barabási

The configuration model starts by creating the n nodes: node i has k_i available “edge stubs”. Then, we keep selecting randomly two available stubs and connect them together with an edge, until there are no available stubs. The process is guaranteed to cover all stubs as long as the sum of all node degrees is even.

The configuration model process is random and so it creates different networks each time, allowing us to produce an ensemble of networks with the given degree distribution.

Additionally, note that the constructed edges may form self-loops (*connecting a node to itself*) or multi-edges (*connecting the same pair of nodes multiple times*). In some applications multi-edges and self-loops are not allowed – but the good news is

that they are unlikely to happen when n is very large and the network is sparse. Removing them artificially is another option but it can cause deviations from the desired degree distribution.

The visualization of this page shows three different networks with $n=4$ that can result from the configuration model, given the same degree distribution. Note that there are more than three different networks that could include self loops and multi-edges - can you find the rest?

Food for Thought

What is the probability that the configuration model will connect two nodes of degree k_i and k_j ?